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## PAPER\_997: Production Scaling v13 — 550k calc/s

### Abstract

We upgrade the production benchmark from v11 (500k calc/s, 16 kernels) to v13 (550k calc/s, 20 kernels). Four new kernels are added:

| Kernel                    | Description                                       |
|---------------------------|---------------------------------------------------|
| kernel_fubi_inside_out    | F_U_Bi inside-to-outside mass portion             |
| kernel_99sys_gamma_sweep  | 99-system aggregate at given $\Gamma$             |
| kernel_agn_cena_fubi      | Centaurus A AGN F_U_Bi_i with jet modulation      |
| kernel_ns_merger_gw190425 | GW190425 NS merger strain with phonon suppression |

### 1. Scaling History

| Version    | Target (calc/s) | Kernels   |
|------------|-----------------|-----------|
| v4         | 100,000         | 4         |
| v7         | 300,000         | 8         |
| v11        | 500,000         | 16        |
| <b>v13</b> | <b>550,000</b>  | <b>20</b> |

### 2. Benchmark Architecture

```
class ProductionScalingV13:
    TARGET = 550_000
    def single_pass(self) -> float:      # Execute all 20 kernels
    def simulate(self, duration_s)        # Timed benchmark loop
    def compute(self, dataset)            # Full pipeline with metadata
```

### 3. Implementation

File: production\_scaling\_v13.py, class ProductionScalingV13. CP4 class #581.